

Tutorial sheet - 06

Section - A

Q.1 Define Power series.

⇒ A power series is of the form $f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$
or in general form $f(x) = \sum_{n=0}^{\infty} a_n x^{n+1}$

Q.2 Define ordinary point

⇒ let us consider linear differential eqⁿ $P_0 \frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_2 y = 0$

where P_0, P_1, P_2 are polynomial in x , a point $x=0$ is ordinary if $P_0(x) \neq 0$ for $x=0$.

Q.3 Define singular point

⇒ A point $x=0$ is said to be singular point if $P_0(x) = 0$ at $x=0$ for the eqⁿ $P_0 \frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_2 y = 0$ when $\frac{P_1}{P_0}$ and $\frac{P_2}{P_0} = \infty$ at $x=0$

$$\text{Ex} \rightarrow x^2 \frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} + xy = 0$$

Q.4 Define Bessel's equation.

⇒ A differential equation of the form $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$ is called Bessel's equation.

$$\text{E.g :- } x^2 y'' + xy' + (x^2 - 4)y = 0$$

Q.5 Define Legendre's equation

⇒ The diff. equation of the form $(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$ is called Legendre's equation

E.g:- $(1-x^2)y'' - 2xy' + 6y = 0$

PART - B

Q.1 $(1-x^2)y'' - 2xy' + 2y = 0$ — (1)

⇒ $y = \sum_{n=0}^{\infty} a_n x^n$ (let) — (2)
 $y' = \sum_{n=0}^{\infty} n x^{n-1} a_n$, $y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$

Put y', y, y'' in eqⁿ (1)

$$(1-x^2) \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} - 2x \sum_{n=0}^{\infty} n x^{n-1} a_n + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=0}^{\infty} n(n-1) a_n x^n - 2 \sum_{n=0}^{\infty} n x^n a_n + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} a_n n(n-1) x^{n-2} - \sum_{n=0}^{\infty} [n(n-1) + 2n - 2] a_n x^n = 0$$

↓
Put $n=2$

↓
Put $n=0$

$$a_2 \cdot 2(2-1) - [-2]a_0 = 0$$

$$2a_2 + 2a_0 = 0$$

$$a_2 = -a_0$$

Put $n = n+2$ in first term and $n = n$ in second $a_{n+2}(n+2)(n+1) - [n(n-1) + 2n - 2] a_n = 0$

$$a_{n+2}(n+2)(n+1) - (n-1)(n+2)a_n = 0$$

$$a_{n+2} = \frac{n-1}{n+1} a_n$$

$$\left. \begin{aligned} a_2 &= -a_0 \quad (n=0) \\ a_4 &= -\frac{1}{3} a_0 \quad (n=2) \\ a_6 &= -\frac{1}{6} a_0 \quad (n=4) \end{aligned} \right\} \begin{aligned} a_1 &= 0 \quad (n=1) \\ a_3 &= 0 \quad (n=3) \\ \text{odd terms} &= 0 \end{aligned}$$

from eqn (2), $y = \sum_{n=0}^{\infty} a_n x^n \Rightarrow a_0 + a_1 x + a_2 x^2 + a_3 x^3 \dots$

$$y = a_0 + 0 + (-a_0)x^2 + 0 - \frac{1}{3}a_0 x^4 + 0 - \frac{x^6}{5}a_0 + a_1 x$$

$$y = a_0 \left[1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} + \dots \right] + a_1 x$$

Q.2 Solve $(1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$ — (1)

$\Rightarrow y = \sum_{n=0}^{\infty} a_n x^n$ (let) — (2)

$$y' = \sum_{n=0}^{\infty} a_n \cdot n \cdot x^{n-1}, \quad y'' = \sum_{n=0}^{\infty} a_n \cdot (n)(n-1) x^{n-2}$$

Put y, y', y'' in eqn (1)

$$(1+x^2) \sum_{n=0}^{\infty} a_n \cdot n \cdot (n-1) x^{n-2} + x \sum_{n=0}^{\infty} a_n \cdot n \cdot x^{n-1} - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} a_n \cdot n \cdot (n-1) x^{n-2} + \sum_{n=0}^{\infty} a_n \cdot n \cdot (n-1) x^n + \sum_{n=0}^{\infty} a_n \cdot n \cdot x^n - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} a_n \cdot n(n-1) x^{n-2} + \sum_{n=0}^{\infty} a_n [n(n-1) + n - 1] x^n = 0$$

Put $n = n+2$

Put $n = n$

$$a_{n+2} \cdot (n+2)(n+1) + a_n [n^2 - 1] = 0$$

$$a_{n+2} = \frac{-[n^2-1]}{(n+2)(n+1)} a_n = \frac{(1-n)}{n+2} a_n$$

$$n=0 \rightarrow a_2 = \frac{1}{2} a_0$$

$$n=2 \rightarrow a_4 = \frac{-1}{4} a_2 \Rightarrow \frac{-1}{8} a_0$$

$$n=1 \rightarrow a_3 = 0$$

$$n=3 \rightarrow a_5 = 0$$

higher terms = 0

from eqⁿ ②, $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$

$$y = a_0 + a_1 x + \frac{1}{2} a_0 x^2 + 0 + \frac{-1}{8} a_0 x^4 + \dots$$

$$y = a_0 \left[1 + \frac{x^2}{2} - \frac{x^4}{8} \right] + a_1 x$$

Q.3 $y'' + xy' - x^2 y = 0$ — ①

$\Rightarrow y = \sum_{n=0}^{\infty} a_n x^n$ (let) — ②

$$y' = \sum_{n=0}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$$

put y, y', y'' in eqⁿ ①

$$\sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

put $n = n+2$

put $n = n$

put $n = n-2$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} a_{n-2} x^n = 0$$

$$(n+2)(n+1)a_{n+2} + (n)a_n - a_{n-2}$$

$$a_{n+2} = \frac{a_{n-2} - n \cdot a_n}{(n+2)(n+1)}$$

$$n=0 \rightarrow a_2 = \frac{a_{-2} - 0}{2}$$

(a_{-2} does not exist)

$$a_2 = 0$$

$$n=2 \rightarrow a_4 = \frac{a_0 - 0}{12}$$

$$n=1 \rightarrow a_3 = \frac{a_{-1} - a_1}{6}$$

(a_{-1} does not exist)

$$a_3 = -\frac{a_1}{6}$$

$$n=3 \rightarrow a_5 = \frac{a_1 - 3a_3}{20} = \frac{3a_1}{40}$$

from eqⁿ (2) $\rightarrow y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$

$$y = a_0 + a_1 x + 0 + \frac{a_1}{6} x^3 + \frac{a_1}{12} x^4 + \frac{3a_1}{40} x^5 + \dots$$

$$y = a_0 \left[1 + \frac{x^4}{12} \dots \right] + a_1 \left[x - \frac{x^3}{6} + \frac{3x^5}{40} \dots \right]$$

PART - C

Q.1 Solve $\frac{d^2 y}{dx^2} + x^2 y = 0$ — (1)

$\Rightarrow$ Let $y = \sum_{n=0}^{\infty} a_n x^n$ — (2)

$$y' = \sum_{n=0}^{\infty} n \cdot a_n x^{n-1}, y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} = 0$$

Put y, y'' in eqⁿ (1)

$$\sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

↓
put $n = n+2$

↓
put $n = n-2$

$$(n+2)(n+1) a_{n+2} + a_{n-2} = 0 \Rightarrow a_{n+2} = \frac{-a_{n-2}}{(n+2)(n+1)}$$

$$n=0 \rightarrow a_2 = \frac{-a_{-2}}{2}$$

(a_{-2} does not exist)

$$a_2 = 0$$

for $n=2$

$$a_4 = \frac{-a_0}{12}$$

for $n=4$

$$a_6 = \frac{-a_2}{30} = 0$$

for $n=6$

$$a_8 = \frac{-a_4}{8 \cdot 7} \Rightarrow \frac{a_0}{8 \cdot 7 \cdot 12}$$

for $n=8$

$$a_{10} = 0$$

for $n=10$

$$a_{12} \Rightarrow \frac{-a_8}{12 \cdot 11} \Rightarrow \frac{-a_0}{12 \cdot 11 \cdot 8 \cdot 7 \cdot 12}$$

from eqⁿ (2)

$$n=1 \rightarrow a_3 = \frac{-a_{-1}}{6}$$

(a_{-1} does not exist)

$$a_3 = 0$$

for $n=3$

$$a_5 = \frac{-a_1}{20}$$

for $n=5$

$$a_7 = 0 \quad (\text{as } a_3 = 0)$$

for $n=7$

$$a_9 = \frac{-a_5}{9 \cdot 8} \Rightarrow \frac{a_1}{20 \cdot 8 \cdot 9}$$

$$y = \sum_{n=0}^{\infty} a_n x^n \Rightarrow a_0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + a_4 x^4 \dots$$

$$y = a_0 + a_1 x + 0 + 0 - \frac{a_0 x^4}{12} - \frac{a_1 x^5}{20} + 0 + 0 + \frac{a_0}{8 \cdot 7 \cdot 12} \dots$$

$$y = a_0 \left[1 - \frac{x^4}{12} + \frac{x^8}{12 \cdot 7 \cdot 8} - \frac{x^{12}}{12 \cdot 7 \cdot 8 \cdot 11 \cdot 12} \dots \right] + a_1 \left[x - \frac{x^5}{20} + \frac{x^9}{20 \cdot 8 \cdot 9} \dots \right]$$

Q.2 Solve Bessel's Equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0$ — (1)

$\Rightarrow$ Let $y = \sum_{r=0}^{\infty} a_r x^{m+r}$ — (2)

$$y' = \sum_{r=0}^{\infty} (m+r) a_r x^{m+r-1}, \quad y'' = \sum_{r=0}^{\infty} (m+r)(m+r-1) a_r x^{m+r-2}$$

put y, y', y'' in eqⁿ (1)

$$x^2 \sum_{r=0}^{\infty} (m+r)(m+r-1) a_r x^{m+r-2} + x \sum_{r=0}^{\infty} (m+r) a_r x^{m+r-1} + (x^2 - n^2) \sum_{r=0}^{\infty} a_r x^{m+r} = 0$$

$$\sum_{r=0}^{\infty} [(m+r)(m+r-1) + (m+r) - n^2] a_r x^{m+r} + \sum_{r=0}^{\infty} a_r x^{m+r+2} = 0$$

$$\sum_{r=0}^{\infty} [m^2 + 2mr - m + r^2 - 1 + m + 1 - n^2] a_r x^{m+r} + \sum_{r=0}^{\infty} a_r x^{m+r+2} = 0$$

$$\sum_{r=0}^{\infty} [(m+r)^2 - n^2] a_r x^{m+r} + \sum_{r=0}^{\infty} a_r x^{m+r+2} = 0$$

put $r=0$

$$(m^2 - n^2) a_0 = 0 \rightarrow a_0 \neq 0$$

$$m = \pm n, \quad m = n$$

put $r=1$ in first

$$[(m+1)^2 - n^2] a_1 = 0$$

$$m+1^2 - n^2 \neq 0 \Rightarrow \boxed{a_1 = 0}$$

Equating coefficients $\rightarrow$ put $r = r+2$ in first term

$$[(m+r+2)^2 - n^2] a_{r+2} + a_r = 0$$

$$a_{n+2} = \frac{-a_n}{(m+n+2)^2 - n^2}$$

for $n=0 \rightarrow a_2 = \frac{-a_0}{(m+2)^2 - n^2}$, for $n=1 \rightarrow a_3 = \frac{-a_1}{(m+3)^2 - n^2}$, all odd terms are zero

for $n=2 \rightarrow a_4 = \frac{-a_2}{(m+4)^2 - n^2} \Rightarrow \frac{a_0}{[(m+2)^2 - n^2][(m+4)^2 - n^2]}$

from eqn (2) $y = \sum_{n=0}^{\infty} a_n x^{n+n} = a_0 x^n + a_1 x^{n+1} + a_2 x^{n+2} + a_3 x^{n+3} \dots$

$$y = a_0 x^n + 0 + \frac{a_0 x^{n+2}}{[(m+2)^2 - n^2]} + \frac{a_0 x^{n+4}}{[(m+2)^2 - n^2][(m+4)^2 - n^2]} \dots$$

for $m=n$

$$y = a_0 \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right] x^n$$

for $m=-n$

$$y = a_0 \left[1 - \frac{x^2}{2(2-2n)} + \frac{x^4}{2 \cdot 4(2-2n)(4-2n)} - \dots \right] x^{-n}$$